

ELEMENTS OF COMPUTER SCIENCE AND ENGINEERING

I B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS01	HSMC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	100	0	100

COURSE OBJECTIVES

The course will enable the students to:

1. To provide an overview of the subjects of computer science and engineering.
2. To Know different types of operating system, database systems.
3. To understand the significance of networks, internet, and WWW and cyber security.
4. To understand Autonomous systems, the application of artificial intelligence.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Know the working principles of functional units of a basic Computer
2. Understand program development, the use of data structures and algorithms in problem solving.
3. Know the need and types of operating system, database systems.
4. Understand the significance of networks, internet, and WWW and cyber security.
5. Understand Autonomous systems, the application of artificial intelligence.

UNIT - I	BASICS OF A COMPUTER	CLASSES: 10
Basics of a Computer – Hardware, Software, Generations of computers. Hardware - functional units, Components of CPU, Memory – hierarchy, types of memory, Input and output devices. Software –systems software, application software, packages, frameworks, IDEs.		
UNIT - II	SOFTWARE DEVELOPMENT	CLASSES: 10
Software development – waterfall model, Agile, Types of computer languages – Programming, markup, scripting Program Development – steps in program development, flowcharts, algorithms, data structures – definition, types of data structures		
UNIT - III	OPERATING SYSTEMS AND DATABASE MANAGEMENT SYSTEMS	CLASSES: 10
Operating systems: Functions of operating systems, types of operating systems, Device & Resource management Database Management Systems: Data models, RDBMS, SQL, Database Transactions, data centers, cloud services		
UNIT - IV	COMPUTER NETWORKS	CLASSES: 10
Computer Networks: Advantages of computer networks, LAN, WAN, MAN, internet, Wi-Fi, sensor networks, vehicular networks, 5G communication. World Wide Web – Basics, role of HTML, CSS, XML, Tools for web designing, Social media, Online social networks. Security – information security, cyber security, cyber laws		

UNIT - V	AUTONOMOUS SYSTEMS	CLASSES: 10
Autonomous Systems: IoT, Robotics, Drones, Artificial Intelligence – Learning, Game Development, natural language processing, image and video processing. Cloud Basics		
TEXT BOOKS		
1. Invitation to Computer Science, G. Michael Schneider, Macalester College, Judith L. Gersting University of Hawaii, Hilo, Contributing author: Keith Miller University of Illinois, Springfield. 5 th Edition, 2010		
REFERENCE BOOKS		
1. Fundamentals of Computers, Reema Thareja, Oxford Higher Education, Oxford University Press, 2 nd Edition, 2020.		
2. Introduction to computers, Peter Norton, 8th Edition, Tata McGraw Hill.		
3. Computer Fundamentals, Anita Goel, Pearson Education India, 2010.		
4. Elements of computer science, Cengage		